

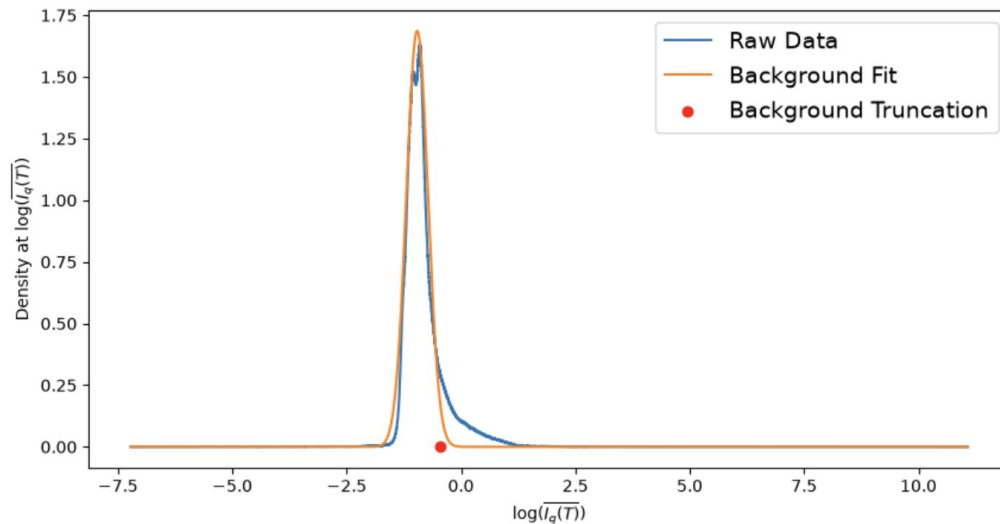
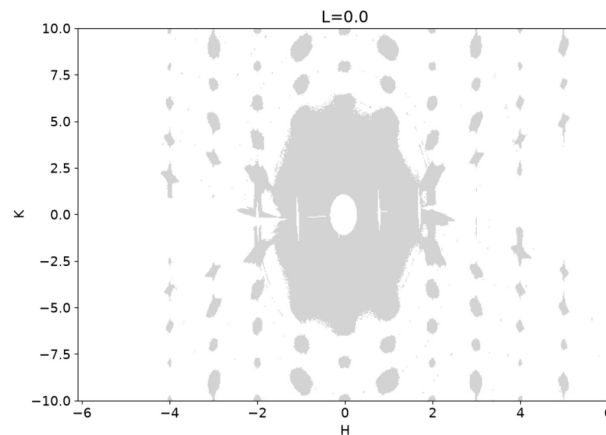
Continuing Advanced Clustering

Goals For This Week

- Re-plot Log Intensity Distribution with our data
- Automated selections of parameters
 - Amount of clusters and rescaling method
- CPU/GPU comparison, time analysis

Log Intensity Distribution

- Pixel intensity is averaged over 24 temperatures, then plotted
- The cut off is the red dot/background truncation
- Pixels brighter than the cut off are kept (the rest is noise)
- Q-space map shows Bragg peaks and air scattering



Automated Parameter Selection Using Sweep and Judge

Sweep runs clustering across every combination of parameters

- Number of clusters
- XTEC-d or XTEC-s
- Initialization method
 - K-Means++
- Rescale method (one per run)
 - Mean, z-score, log-mean, none

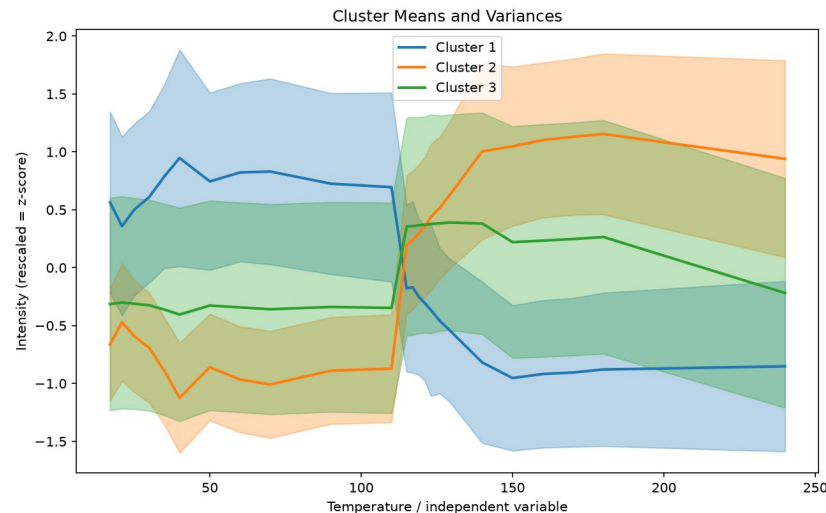
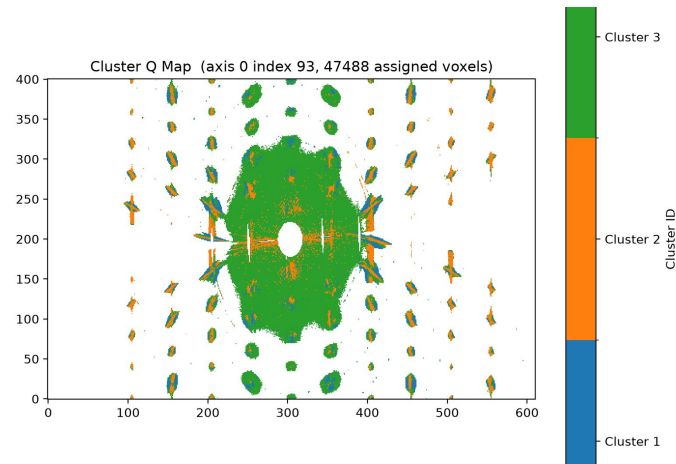


Judge scores each result on six criteria (higher score is better)

- BIC - how well the model fits
- Trajectory separation - how distinct cluster curves are
- Spatial coherence - do neighboring pixels share a cluster
- Trajectory smoothness - real temperatures should be smooth
- Size penalty - flags tiny clusters
- Jump penalty - flags clusters with sudden spikes

Sweep and Judge Results

- Best number of clusters: $k=3$
- Best rescale method: z-score
 - Coherence and size penalties
- Clean, distinct trajectories
- The judge ranked the options, but its top pick was misleading
- I used my own judgement, scale-free metrics, and trajectory shape to find the trustworthy answer
- This is why an AI agent would help



CPU/GPU Time Comparison

- XTEC-GPU is built to run on a GPU, but QM2 doesn't have easy access to a GPU
- CPU vs GPU (benchmark from XTEC-GPU repo on their data, not ours)
 - Wall time:
 - CPU 55.37s
 - GPU 12.17s
 - Clustering phase:
 - CPU 46.20s
 - GPU 2.55s

Goals For Next Week

Build an AI agent to automate clustering decision

- How should I implement?
- How should it run?
- What are the goals?
 - Come up with a combination of scores that would allows us to rank different hyperparameter choices